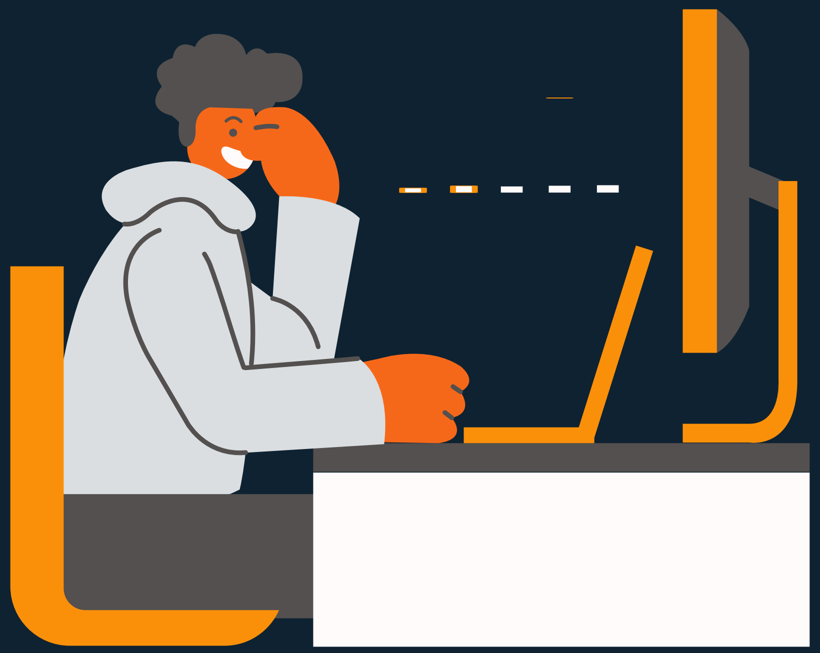
FINANCIAL INFORMATION RETRIEVAL FOR OPINION-AWARE DOCUMENT RANKING



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Oliwia Iwańska

Chiara Pierini

Marta Paniconi

WORK ORGANIZATION

- **Joint problem formulation and dataset analysis**
- **Baseline implementation and exploratory experiments**
- **Neural and hybrid retrieval design**
- **Evaluation, comparison, and reporting**

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PROBLEM DESCRIPTION

1 **Why this task matters?**

- **~36%** of FiQA queries contain **opinion, sentiment, or uncertainty cues.**
- Users seek explanations, viewpoints, and sentiment, not only facts
- Financial decisions are influenced by **market sentiment and collective opinion**

2 **Key characteristics:**

- Short, informal, and subjective financial texts
- Limited numeric content
- Presence of explicit evaluative and sentiment cues

3 **Why this is interesting:**

Relevance depends on **semantic meaning and sentiment, not only keywords**

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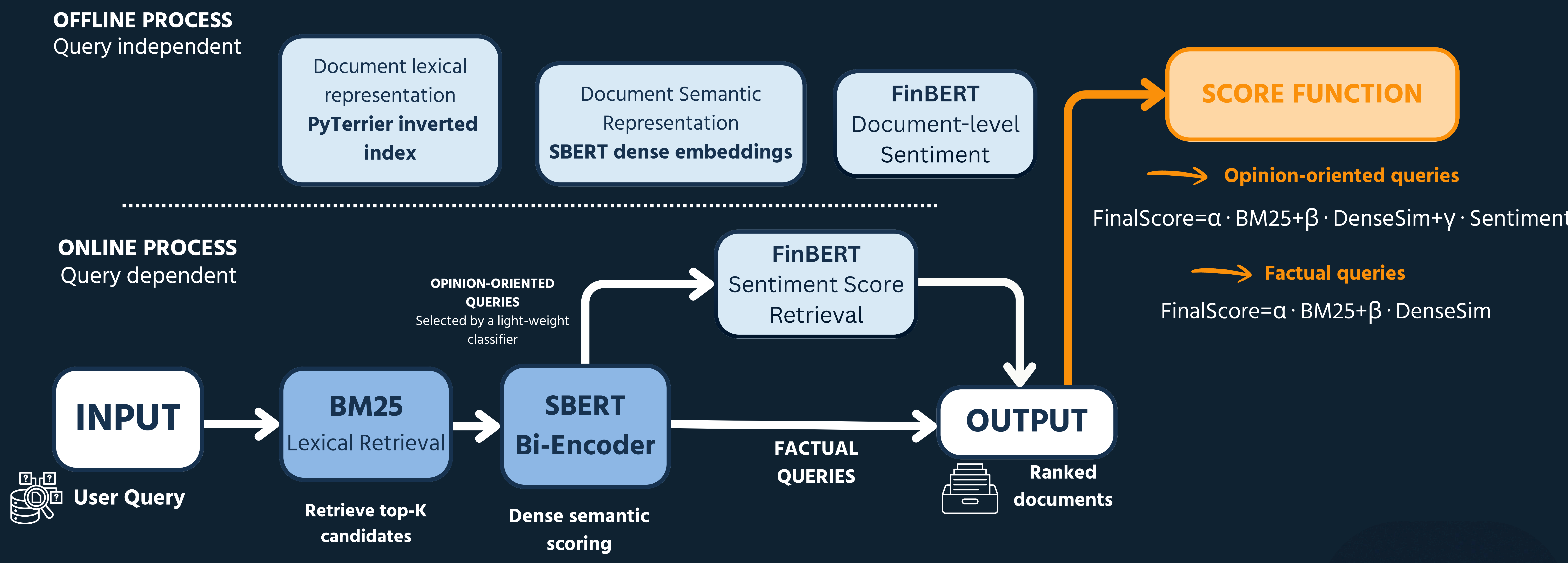
PROPOSED SOLUTION



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VS BASELINES FOR COMPARISON

We compared our approach against three lexical baselines:

1

BM25 (Lexical)
Pure keyword-based retrieval

2

BM25 + RM3 (Query Expansion)
Pseudo-relevance feedback for query expansion

3

BM25 + Entity Matching
Lightweight entity overlap (company/ticker)

Why important

- Establishes a **pure lexical lower bound**
- Shows **limitations** of keyword-only retrieval
- **Reference** point for all improvements

- Addresses **short** and **ambiguous** queries
- Tests whether **expansion** alone is sufficient

- Finance is **entity-centric** (companies, tickers)
- Evaluates early **entity-aware** signals

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EVALUATION SET UP

We evaluate all methods on the FiQA test set using standard IR metrics.

Recall: R@5, R@10

Measures how many relevant documents are retrieved within the top-K results.

Precision: P@1, P@5, P@10

Measures how many of the top-K retrieved documents are actually relevant.

Ranking Quality: nDCG@5, nDCG@10

Evaluates how well relevant documents are ranked, giving more weight to top positions

Overall Effectiveness: MAP

Measures overall retrieval performance across all queries, considering ranking quality.

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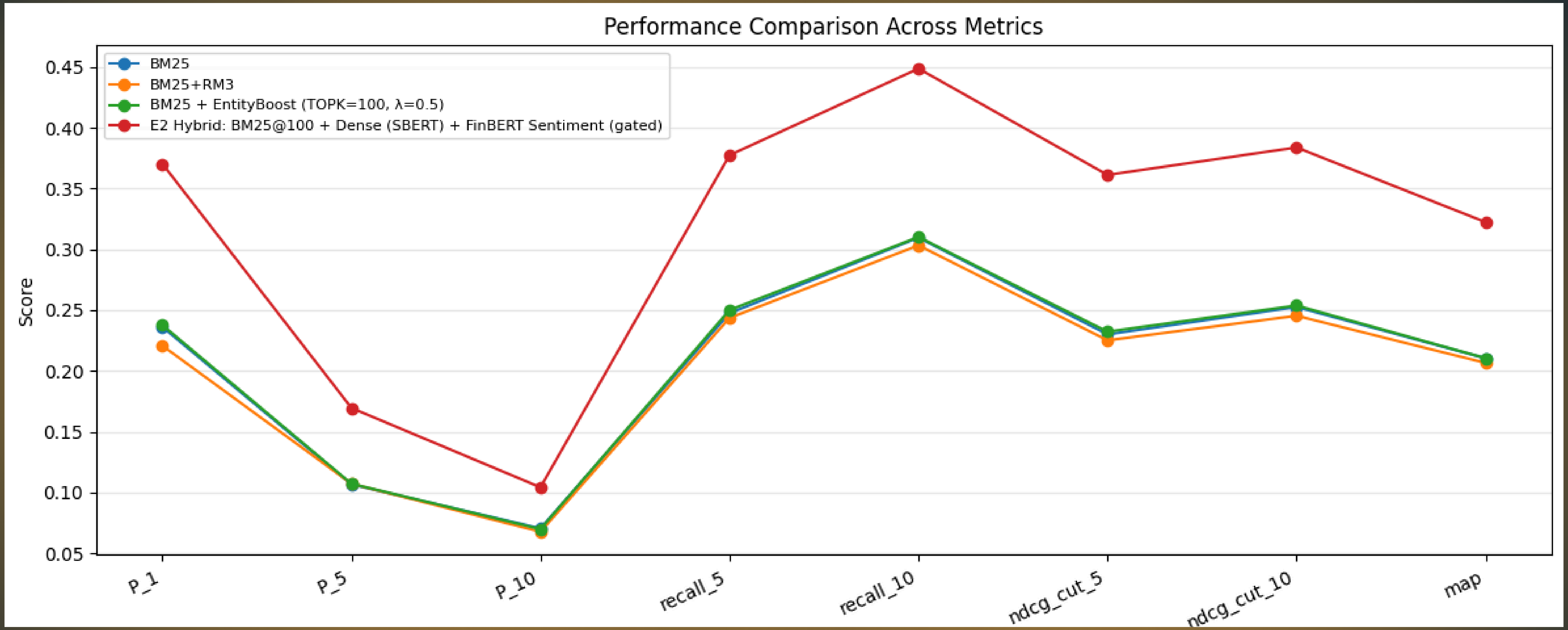
RESULTS



- Lexical matching alone fails to capture evaluative intent.
- RM3 query expansion introduces noise for short opinion queries.
- Simple entity matching yields only marginal gains.
- Hybrid lexical–semantic retrieval improves robustness.
- Selective sentiment weighting refines opinion-oriented rankings.

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LESSON LEARNED

Effective financial IR systems require a careful balance of lexical, semantic, and opinion-aware components

Separating offline and online computation improves efficiency and scalability

Dataset analysis is critical for guiding model design choices

Baseline experiments and E1 were essential for identifying which components will truly improved performance

Simply adding more components does not guarantee better performance, careful weighting is essential

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LIMITATIONS

in our approach

- Sentiment modeling is noisy and only helpful for some queries.
- Query intent detection is heuristic-based rather than learned.
- Dense retrieval may retrieve semantically related but not strictly relevant documents.

in experimental setting

- Binary relevance judgments limit fine-grained evaluation.
- Sparse relevance (few relevant documents per query) makes evaluation difficult.
- Experiments are conducted on a single dataset (FiQA).

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AI TOOLS

AI tools used

- **ChatGPT 5.2** (assistive support for writing and clarification)

Role of AI

- AI tool was used only as support
- All system design, implementation, experiments, and evaluations were conducted by the authors
- No AI tool was used to generate results or replace critical reasoning

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THANK YOU FOR YOUR ATTENTION.

WE ARE HAPPY TO TAKE QUESTIONS!

A.Y. 2025-2026

Marta Paniconi-535358
Oliwia Iwanska- 535418
Chiara Pierini-535407

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